

MedS³: Towards Medical Slow Thinking with Self-Evolved Soft Dual-sided Process Supervision

Shuyang Jiang^{1,3}, Yusheng Liao^{2,3}, Zhe Chen^{2,3}, Ya Zhang^{2,3*}, Yanfeng Wang^{2,3}, Yu Wang^{2,3*}

¹Fudan University

²Shanghai Jiao Tong University

³Shanghai Artificial Intelligence Laboratory

shuyangjiang23@m.fudan.edu.cn

{liao20160907,chenzhe2018,ya_zhang,wangyanfeng622,yuwangshtu}@sjtu.edu.cn

Abstract

Medical language models face critical barriers to real-world clinical reasoning applications. However, mainstream efforts, which fall short in task coverage, lack fine-grained supervision for intermediate reasoning steps, and rely on proprietary systems, are still far from a versatile, credible and efficient language model for clinical reasoning usage. To this end, we propose MedS³, a self-evolving framework that imparts robust reasoning capabilities to small, deployable models. Starting with 8,000 curated instances sampled via a curriculum strategy across five medical domains and 16 datasets, we use a small base policy model to conduct Monte Carlo Tree Search (MCTS) for constructing rule-verifiable reasoning trajectories. Self-explored reasoning trajectories ranked by node values are used to bootstrap the policy model via reinforcement fine-tuning and preference learning. Moreover, we introduce a soft dual process reward model that incorporates value dynamics: steps that degrade node value are penalized, enabling fine-grained identification of reasoning errors even when the final answer is correct. Experiments on eleven benchmarks show that MedS³ outperforms the previous state-of-the-art medical model by +6.45 accuracy points and surpasses 32B-scale general-purpose reasoning models by +8.57 points. Additional empirical analysis further demonstrates that MedS³ achieves robust and faithful reasoning behavior.

1 Introduction

Large Language Models (LLMs) have demonstrated significant potential in the medical domain (Singhal et al. 2023; Nori et al. 2023; Chen et al. 2023), supporting tasks from clinical note generation (Biswas and Talukdar 2024; Jung et al. 2024) to patient communication (Tu et al. 2024; Liao et al. 2024). Despite these advances, accurate reasoning is steadily fundamental to clinical decision-making, where diagnostic and treatment recommendations must be grounded in coherent, evidence-based logic chains (Cabral et al. 2024; Tordjman et al. 2025). Increasing efforts to enhance reasoning capabilities through chain-of-thought (Wei et al. 2022), preference learning (Rafailov et al. 2023) and reinforcement learning (Guo et al. 2025) highlight that correct answers alone are insufficient without trustworthy reasoning processes.

While existing approaches have demonstrated notable performance, two challenges persist. First, the training data used in many studies mostly consist of multiple-choice problems (Huang et al. 2025b,a), which lacks sufficient diversity and scale, and hence limits model robustness across different domains. Second, a growing dependence on large-scale proprietary models (Huang et al. 2025b) introduces practical and ethical considerations. Although models distilled from these hyper-scale teachers achieve strong performance, they inherit the unverifiability and potential hallucinations (Xu, Jain, and Kankanhalli 2024) of their teachers, offering little control over reasoning faithfulness. Moreover, the reliance on distillation from external resources would lead to uncontrollable privacy protection for real-world applications. These challenges highlight a core problem: how to efficiently induce robust, interpretable, and stepwise supervision reasoning in small-scale medical models without relying on proprietary models or noisy synthetic supervision.

To bridge this gap, we propose MedS³, a self-evolving medical reasoning framework that enables small models to iteratively improve through Monte Carlo Tree Search (MCTS)-guided exploration and rule-verified refinement. Starting from a diverse, curriculum-sampled dataset spanning five medical domains and 16 medical datasets, MedS³ generates reasoning trajectories with explicit node value estimates, allowing selection of high-quality paths for policy model bootstrapping via reinforcement fine-tuning and preference learning. Crucially, we introduce a soft dual-sided process reward model that labels intermediate steps not only by potential correctness but also by value consistency—penalizing steps that degrade node value and marking them as incorrect if the adjusted value falls below zero. This enables faithful supervision even in trajectories with correct final answers. Table 1 highlights these advantages in robust long-chain reasoning and breadth of application.

Extensive experiments on eleven clinical reasoning benchmarks and three out-of-domain datasets demonstrate that MedS³ achieves state-of-the-art performance, outperforming both comparable-sized medical models and much larger general reasoning models, while maintaining superior interpretability and clinical task coverage. In summary, our contributions are:

*Corresponding Author

Models	Without Close-sourced Teacher	Diverse Clinical Coverage	Small Size	Reasoning Specialized	Process Supervision
UltraMedical	✗	✓	✓	✗	✗
HuatuoGPT-o1	✗	✗	✓	✓	✗
O1-journey Part 3	✗	✗	✗	✓	✗
m1-7B-32K	✗	✗	✓	✓	✗
MedS ³	✓	✓	✓	✓	✓

Table 1: Comparison of MedS³ with other medical models. Our MedS³ supports flexible inference-time scaling on resource-constrained devices, as well as process reward-guided decoding algorithms. Furthermore, MedS³ is a self-evolved model without dependence on large proprietary models for distillation or critique.

1. **Pioneering Step-Level Framework for Medical AI:** We introduce a self-evolution framework that equips small-scale medical models with robust long-chain reasoning via step-level supervision, tailored for a wide range of clinical applications.
2. **Novel PRM Training Pipeline:** We propose a unique process reward model trained with soft dual-sided labels, which precisely evaluates each reasoning step by jointly predicting future rewards and assessing atomic step necessity, reflecting clinical reasoning’s incremental confidence building and fewer hallucinations.
3. **State-of-the-Art Clinical Reasoning Performance:** Our self-evolved system MedS³ significantly surpasses all equal-parameter competitors and larger reasoning models across multiple clinical benchmarks, driven by fine-grained PRM-guided reasoning enhancement.

2 MedS³

This section presents a detailed overview of the proposed MedS³ framework, which is structured into four components:

1. **Self-Bootstrapping Evolution** (§2.1) which synthesizes reasoning trajectories as training data, with Monte-Carlo Tree Search (MCTS) technique using the base policy π_0 .
2. **Policy Model** π (§2.2) which is derived by fine-tuning on the generated synthetic data with supervised learning and direct preference optimization (Rafailov et al. 2023).
3. **Process Reward Model (PRM)** V_θ (§2.3) which is fine-tuned with step-wise supervision using soft dual-side labels and assigns a value in the range $[0, 1]$ to each reasoning step by a both forward and backward view.
4. **Iterative Training Pipeline** (§2.4) which consists of two MCTS evolution iterations and a curriculum data sampler.

The overall framework is presented in Fig. 1.

2.1 MCTS-guided Evolution

This algorithm builds upon an n -ary tree, where every root node is initialized as a reasoning start s_0 = “Let’s break down this problem step by step.” to guarantee a multi-step reasoning process. There are four stages in a full MCTS pipeline, including *Node Selection*, *Node Expansion*, *Node Rollout*, and *Backpropagation*.

Node Selection Within each iteration, we use UCB (Winands, Björnsson, and Saito 2008) as the criterion to select a child T , which is as follows:

$$\text{UCB}_T = v_C + \gamma \sqrt{\frac{\ln n_{T_{\text{parent}}}}{n_T}}, \quad (1)$$

where T_{parent} is the preceding node of the current node T , n_T is the node visiting count, v_C is the node’s value obtained by node rollout and updated by back-propagation, and γ is an exploration constant set as 2. For each parent, we select its child node with the highest UCB value.

Node Expansion After reaching the candidate node T_c under the UCB criterion, we continue the reasoning trace of the current node. If the current node possesses a relatively high value ($v_C \geq \text{thr}$, where $\text{thr} = 0.9$ is a pre-defined threshold), we prompt the node to directly generate until deriving an answer for speeding up exploration. For a wrong node, we allow one reflective action `Reflect` to elicit the introspection of the policy. Otherwise, assume that the selected node is located at k -th layer of the tree with previous reasoning trajectories $[s_0, s_1, \dots, s_k]$ connected by a coherence phrase t_s , we sample B subsequent steps $\{s_{k+1,i} \mid i = 1, 2, \dots, B\}$ based on the previous trajectory using a `Reason`¹ node:

$$s_{k+1,i} \sim \pi_0([s_0 \oplus s_1 \oplus \dots \oplus s_k] \mid x), \quad (2)$$

where $\oplus$ is the operation to connect two steps using the coherence phrase t_s , π_0 is the base policy model, and x is the original input prompt.

Node Rollout As the PRM is not yet accurate enough to serve as a reliable critic, node values are obtained using rollouts based on reasoning trajectories so far. Specifically, for a chosen unvisited node T_c at the k -th depth, we set a simulation budget $L = \min(L_{\min}, \frac{L_0}{k})$, to encourage sufficient simulation trials when the known reasoning path is short, but expect to see a deterministic reasoning result conditioning on a long trajectory. After setting the budget, we prompt the policy model π_0 to directly output the answer L times under a specific prompt `AnsPrompt`:

$$a_c^L \sim \pi_0([s_0 \oplus s_1 \oplus \dots \oplus s_k] \mid x_{\text{AnsPrompt}}), \quad (3)$$

¹Prompts of `Finish`, `Reflect`, and `Reason` actions are illustrated in Appendix E

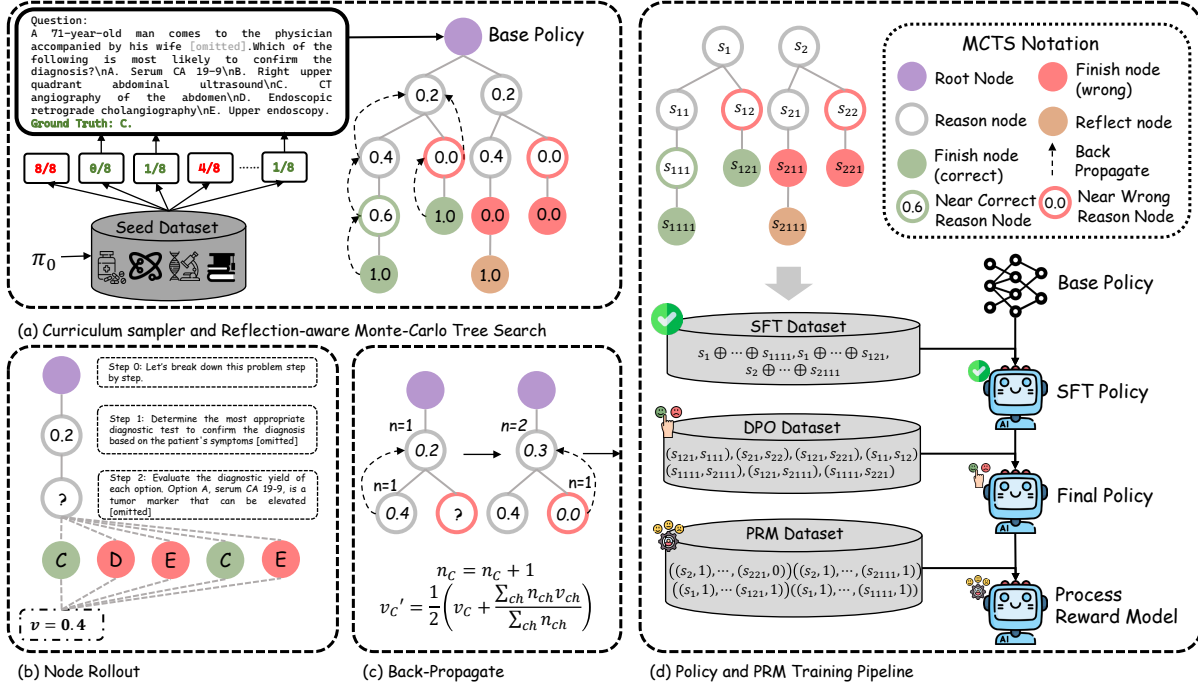


Figure 1: Overview of the construction of MedS³ framework. (a) MedS³ utilizes a Monte-Carlo Tree Search pipeline to self-generate step-by-step reasoning paths for each instance sampled in a curriculum manner. (b) During this process, MedS³ uses result simulation to obtain the rollout value for each node; (c) After obtaining the child’s rollout value, MedS³ executes back-propagation to enable precise value prediction from deeper layers to transfer back to shallow nodes. (d) After the exploration finishes, we use SFT and DPO to optimize the policy model and soft dual-side label to fine-tune the process reward model.

where $l \in [1, L]$ and a_c^l is the l -th simulated answer. The average accuracy of the L simulations $acc = \frac{1}{L} \sum_{l=1}^L \mathbb{1}_{a_c^l = y}$ is assigned as the value of T_c .

Backpropagation After the rollout stage, we conduct back-propagation starting from T_c till the root, updating all tree node values along the trace. Specifically, for an arbitrary node T_k , we propose to update its visits n_k and v_k as follows:

$$\begin{aligned} n_k &= n_k + 1 \\ v_k &= \frac{1}{2} \left(v_k + \frac{\sum_{ch} v_{ch} \cdot n_{ch}}{\sum_{ch} n_{ch}} \right), \end{aligned} \quad (4)$$

which considers both correctness and completeness for the evaluation of a reasoning step.

Termination of Search To balance the exploration cost and optimization of policy and reward models, we set two criteria to terminate the exploration. First, once the total correct count in the tree exceeds a minimum correct count $\tau = 3$, we stop the exploration of this tree. Second, if there are no correct nodes after affording a certain number of node exploration trials, we prompt π_0 to generate Finish node for all leaves.

2.2 Policy Model Fine-tuning

The policy training first leverages the correct leaves s_l whose values equal 1 and corresponding reasoning trajectories gathered before: $D_\pi = \{(x, [s_0 \oplus s_1 \oplus \dots \oplus s_l]) \mid v_l = 1\}$.

These correct reasoning traces are fine-tuned to deduce a self-improved policy model:

$$\mathcal{L}_\pi = -\mathbb{E}_{(x,y) \sim D_\pi} \log p_\theta(y \mid x), \quad (5)$$

where $y = [s_0 \oplus s_1 \oplus \dots \oplus s_l]$ is the whole trajectory. For the second iteration, we further add a step-level Direct Preference Optimization (DPO) to optimize the policy at the same reasoning budget:

$$\begin{aligned} \mathcal{L}_{\text{DPO}} &= -\mathbb{E}_{(x, P^+, P^-) \sim D_{\text{DPO}}} \log \sigma(r_\theta(x, P^+) \\ &\quad - r_\theta(x, P^-)), \end{aligned} \quad (6)$$

where $r_\theta(x, P) = \beta(\log \pi_\theta(P \mid x) - \log \pi_{ref}(P \mid x))$ is the reward and $D_{\text{DPO}} = \{(x, [s_0 \oplus s_1 \oplus \dots \oplus s_k^+], [s_0 \oplus s_1 \oplus \dots \oplus s_k^-]) \mid v_k^+ > v_k^-\}$. DPO training is crucial for deriving a strong policy and PRM, which is elucidated in Table 4.

2.3 Soft Dual-side PRM Fine-tuning

Dataset Collection We first filter out those trees with only correct or incorrect leaves as these trajectories contain extreme value bias. For a Finish leaf T_l in a valid tree, its reasoning trace $[(s_1, v_1), (s_2, v_2), \dots, (s_l, v_l)]$ is one training sample, where each reasoning step is concatenated by “Step k:” to form a complete reasoning trajectory. At the end of each reasoning step s_i (typically a $\backslash n \backslash n$ token), the value v_i is used to derive the token label, which is learned by conditioning on all previous steps in an auto-regressive manner. As a result, the PRM training set is

such $D_{V_\theta} = \{(x, [(s_1, v_1), (s_2, v_2), \dots, (s_l, v_l)]) \mid x \in D_{seed} \wedge s_l \text{ is finish}\}$.

Learning objective Instead of fitting the node value (Zhang et al. 2024a) or learning the pair-wise ranking preference (Guan et al. 2025), we choose to use a binary cross-entropy loss to optimize the PRM for its stability. Although Zhang et al. (2025) suggests that the PRM label should be set to True once the rollout score is above zero, we deem that the rollout score as a soft label has a forward-only bias about reasoning correctness. A wrong intermediate step is still possible to derive a correct answer given correct prefixes, but such hallucinations are not what medical reasoning desires. Therefore, a new step is valued highly only when it can both possibly derive a final answer and improve the correctness of the reasoning trajectory deterministically. As a result, we design a dual-side label y_i for step i using its soft Q-value obtained during MCTS as

$$y_i = \begin{cases} \lceil v_i - \beta \cdot \max(0, v_{i-1} - v_{i+1}) \rceil & v_i < v_{i-1} \\ \lceil v_i \rceil & \text{otherwise} \end{cases} \quad (7)$$

This learning objective encourages PRM to simultaneously look ahead and back to judge the current step and penalize intermediate errors except for valid reflection. Based on these, we optimize V_θ using the following loss function:

$$\mathcal{L}_{V_\theta} = \mathbb{E}_{T_k \sim D_{V_\theta}} \sum_{i=1}^k y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i), \quad (8)$$

where $\hat{y}_i$ is the predicted probability of the given step i and β is a hyperparameter set to 1.0 by a simple grid search (details in Appendix C.1). This dual-sided soft-label training, not only prevents the learning of fuzzy labels (rollout value around 0.5) but also learns to judge a misleading step.

2.4 Training Pipeline

We perform two iterations for the seed dataset. For each iteration, we use **curriculum sampler**, which first prompts the policy model to perform the rejected-sampling on the training set, filtering those training instances with all-correct responses to enhance data efficiency. After that, we sample instances with the lowest average accuracy values during the rejected-sampling process, ensuring that the extremely hard problems (0 accuracy score) are no more than one-third of the total samples. After that, we perform MCTS evolution on the seed data and update the policy model. At the end of the second evolution, we further enhance the policy with DPO and train the PRM using the second iteration’s data.

3 Data Statistics

A slow-thinking system in medical scenarios should both excel at exam-level question answering (QA) and handling real-world clinical scenarios, like diagnosis (Tchango et al. 2022), specific disease syndrome (Lab 2020) and drug-related queries (Huynh et al. 2016). However, previous works mainly focused on a simple scenario, with only limited data diversity, especially multiple-choice QA, to train reasoning models. To approximate realistic clinical usage and promote medical reasoning models on a broader range of clinical tasks, we curate

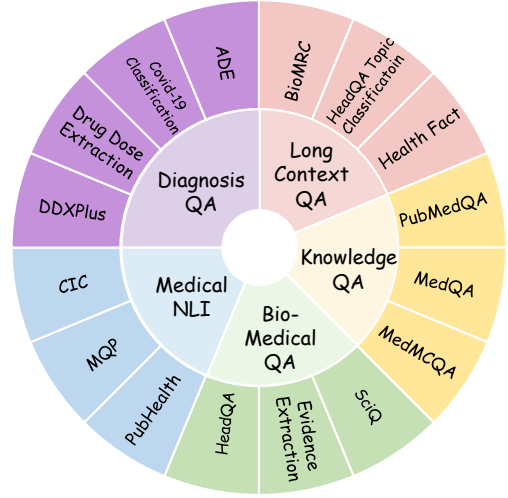


Figure 2: Overview of the used seed datasets.

a training corpus from 16 existing public medical datasets and divide them into five dimensions according to the task category. The five dimensions, i.e., clinical diagnosis QA, natural language inference, knowledge-intensive QA, long-context QA, biomedical QA and corresponding datasets are shown in Fig. 2. Details about the definition of the five dimensions and corresponding tasks are shown in Appendix F.

4 Experiments

In this section, we comprehensively evaluate MedS³ on both in-domain and out-of-domain datasets.

4.1 Experiment Setups

Training and Evaluation We choose Llama3.1-8B-Instruct as the backbone of MedS³. We select MedQA-5op (Jin et al. 2021), PubMedQA (Jin et al. 2019) without contexts, MedMCQA (Pal, Umapathi, and Sankarasubbu 2022), PubHealth (Kotonya and Toni 2020), BioMRC (Pappas et al. 2020), HealFact Classification (Kotonya and Toni 2020), Drug Dose Extraction (Huynh et al. 2016), DDX-Plus (DDX+; Tchango et al. (2022)), the medical subsets of MMLU (Hendrycks et al. 2021), BioASQ (Tsatsaronis et al. 2012) SEER Classification (Dubey et al. 2023) as the evaluation sets. The details of evaluation sets are presented in Appendix F and the hyperparameters of synthesis, self-training and evaluation are presented in Appendix G.

Baselines We choose the following three categories to serve as baselines: (1) LLMs, including GPT-3.5-turbo (OpenAI 2022), GPT-4o-mini (OpenAI 2023), QWQ-preview-32B (Qwen 2024) and R1-Distill-Qwen32B (Guo et al. 2025); (2) Small Language models (<10B), including Llama 3 8B, Llama 3.1 8B (Dubey et al. 2024) and Qwen2.5 7B (Yang et al. 2024), R1-Distill-Llama8B (Guo et al. 2025) (3) Medical LLMs, including MedLlama 3 8B (Yonsei 2024), Med42 (Christophe et al. 2024), OpenBioLLM (Ankit Pal 2024), UltraMedical3-8B and UltraMedical3.1-8B (Zhang

Models	MedQA	MedMCQA	PQA.	BioASQ	MLLU	BioMRC	PubH.	HFact.	DDX+ [†]	DrugD. [†]	SEER [†]	Avg.
<i>Large language models (>10B)</i>												
GPT-4o-mini	75.81	67.58	47.80	83.01	83.79	66.85	59.14	65.24	54.00	73.91	54.54	66.52
GPT-3.5-turbo	59.31	58.12	37.40	74.11	71.11	56.22	57.84	67.85	39.05	86.96	73.61	61.96
QwQ-32B-preview	68.89	61.03	48.60	73.62	74.18	79.76	63.36	66.08	45.40	39.13	37.26	59.76
R1-Distill-Qwen32B	76.83	66.27	38.20	78.32	85.07	78.66	59.95	63.80	53.90	82.61	26.22	64.53
<i>Small language models (<10B)</i>												
Qwen2.5-7B	55.54	54.12	53.40	73.62	74.38	56.48	57.11	52.69	31.25	60.87	33.07	54.78
Llama3-8B	57.50	55.92	56.40	75.73	68.55	56.50	64.09	70.88	35.30	73.91	47.07	60.17
Llama3.1-8B	61.51	57.42	59.00	71.36	72.52	55.60	61.82	63.97	19.00	73.91	52.62	58.98
R1-Distill-Llama8B	50.12	48.89	46.60	70.55	68.42	53.49	55.73	62.04	36.10	69.57	31.71	53.93
<i>Small Medical language models (<10B)</i>												
MedLlama3	55.85	59.36	66.40	<u>84.63</u>	70.08	47.97	62.39	68.10	22.50	69.57	50.69	59.78
Med42	50.20	49.70	55.40	74.76	61.43	57.26	59.14	81.57	31.35	65.22	37.14	56.65
OpenBioLLM	50.20	50.56	41.40	47.73	61.69	27.46	18.77	53.28	16.55	34.78	46.48	40.81
UltraMedical3-8B	68.89	61.82	51.60	80.58	75.08	45.18	66.13	72.73	36.70	60.87	24.55	58.56
UltraMedical3.1-8B	70.93	62.78	56.40	77.18	76.43	54.26	59.14	70.20	31.55	56.52	45.86	60.11
m1-7B-32K	70.70	61.85	48.60	77.83	78.35	52.93	56.70	61.62	29.15	69.57	56.70	60.36
HuatuoGPT-o1	62.53	59.31	58.20	87.70	70.53	50.98	24.61	66.08	40.20	56.52	46.85	56.68
MedS³ (ours)												
Iter 1	65.91	60.55	56.80	78.48	75.66	55.84	57.03	64.73	51.65	73.91	48.97	62.68
Iter 2	67.09	61.56	60.40	80.93	75.21	70.11	68.97	69.87	53.55	91.30	53.44	68.40
Iter 2 w/ PRM	72.97	67.32	64.20	81.39	79.63	74.54	74.41	<u>76.18</u>	62.40	91.30	59.80	73.10

Table 2: Experiment results in 11 in-domain datasets. We highlight the best results with **bold** and underlines the second-best results among models with a similar size. ‘PQA.’ denotes ‘PubMedQA’, ‘PubH.’ denotes ‘PubHealth’, ‘HFact.’ denotes ‘HealthFact’, and ‘DrugD.’ denotes ‘DrugDose’. [†] denotes that the ground truth is not a simple choice index.

Model	MedCalc	MedXpert	RDC
GPT-4o-mini	29.80	15.43	37.80
HuatuoGPT-o1	21.97	16.04	16.20
UltraMedical-3.1-8B	15.19	16.12	21.80
R1-Distilled-Llama-8B	11.94	12.65	13.60
MedS ³	23.69	16.20	33.20
MedS ³ w/ PRM	30.66	16.44	41.20

Table 3: Out-of-domain comparison between MedS³ and previous state-of-the-art models. MedS³ achieves great generalization ability on both policy and process reward models.

et al. 2024b), m1-7B-32K (Huang et al. 2025a) and HuatuoGPT-o1-8B (Chen et al. 2024). All the baselines are evaluated using CoT while MedS³ w/ PRM scores each response with the minimum step value and uses Best-of-N (N=32) to select the final response.

4.2 Main Results

We present the experiment results in Table 2, splitting into examination QA and clinical application tasks. The results unveil that most prior medical LLMs show superior results in traditional multiple-choice problems; while such superiority falls short on out-of-distribution real-world clinical benchmarks (DDXPlus or SEER), resulting in a sub-optimal overall performance compared to the general LLM–Llama3-8B. In contrast, our MedS³ is tailored for universal medical applications and hence achieves the best overall performance among

all open-sourced competitions. As an 8B system, MedS³ achieves +14.12 average performance gains with respect to the base model in the overall assessment, outperforming both medical-oriented models and general reasoning models. After two iterations, the policy model individually achieved the state-of-the-art (SoTA) performance, based on which the soft dual-side PRM further brings an additional 4.7 points improvement. Notably, unlike previous methods that rely on large volumes of multiple-choice queries and consequently suffer from over-fitting, MedS³ achieves robust reasoning improvements, demonstrating that as few as 1,000 high-quality seed examples per task are sufficient to attain superior clinical reasoning performance.

4.3 Generalization to Out-of-domain Tasks

To validate the efficacy of MedS³ on real-world tasks with little labeled data, we select the most frontier models, including GPT-4o-mini, R1-Distill-Llama8B, HuatuoGPT-o1 and UltraMedical3.1-8B as the competitors and further compare MedS³ on MedCalc (Khandekar et al. 2024), MedXpert (Zuo et al. 2025) and the rare disease confirmation (RDC) part sourced from PMCPatients (Zhao et al. 2023). Experiment results in Table 3 illustrate that both the policy and the PRM are applicable to unseen problems and the reasoning manner incentivized by self-evolution is sufficient for both clinical rare disease reasoning and more challenging reasoning scenarios.

Setting	MedQA	MedMCQA	PQA	BioASQ	MMLU	BioMRC	PubH.	HFact.	DDX+	DrugD.	SEER	Avg.
SFT Policy	64.69	61.46	57.80	80.26	75.98	63.28	63.44	64.23	52.65	78.26	48.85	64.63
w/ DPO	67.09	61.56	60.40	80.93	75.21	70.11	68.97	69.87	53.55	91.30	53.44	68.40
w/ H-S label	68.97	65.67	61.80	79.45	76.75	70.48	69.13	74.24	59.35	86.96	56.94	69.98
w/ H-D label	66.77	63.78	61.40	80.74	75.14	78.13	69.54	75.34	61.60	91.30	56.46	70.93
w/ S-D label	72.97	67.32	64.20	81.39	79.63	74.54	74.41	76.18	62.40	91.30	59.80	73.10
w/ SFT init. PRM	70.70	64.40	61.80	81.23	77.39	70.22	75.30	74.58	60.15	82.61	54.99	70.31

Table 4: Ablation study on each component of MedS³ after the second iteration. “H-S” means hard single-sided label, “H-D” means hard dual-sided label, and “S-D” is soft dual-sided label used in MedS³.

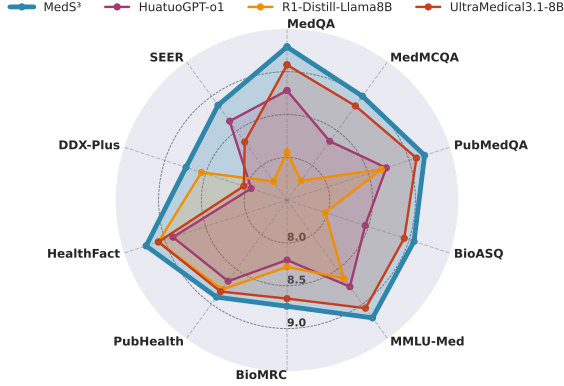


Figure 3: Interpretability evaluation for models using synthetic data, where MedS³ produces the least hallucinatory contents among other pioneering models.

5 Analysis

5.1 Ablation Study

In this section, we validate the effectiveness of each submodule of MedS³. Starting from the SFT-tuned policy model, we compare the final performance with (1) w/ DPO: use DPO to fine-tune the policy; (2) w/ H-S label: conduct best-of-N evaluation using a PRM trained with hard single-sided label (Zhang et al. 2025); (3) w/ H-D label: same as (2) but use hard dual-sided label (Wang et al. 2025) to train a PRM and (4) w/ S-D label (ours): same as (2) but use soft dual-sided label proposed in MedS³ to train a PRM. We also compare with (5) w/ SFT init. PRM, which is the same as (4) but initializes PRM from the SFT-tuned policy, to further show the significance of a PRM exposed to both positive and negative responses. Experiment results in Table 4 show that the DPO helps to greatly improve the policy model, especially in clinical tasks. Furthermore, innovatively determining the dual side label based on the MC estimation, our method is more robust and flexible than rule-based labels, and hence outperforms previous training objectives, confirming the necessity of holistic modeling of a PRM.

5.2 Reliability of MedS³

The non-eliminable hallucinations prevent most medical LLMs from being practical. Albeit inevitability, MedS³ leverages a fine-grained soft dual-sided PRM to improve interpretability and mitigate hallucinatory contents. We lever-

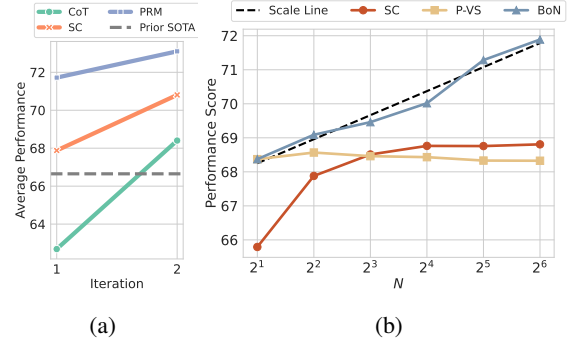


Figure 4: Scaling in (a) self-evolution iterations and (b) sampling numbers during test-time. Both the policy and PRM harvest consistent enhancement with self-evolution, and hence their cooperative system MedS³ achieves a log-linear scaling rate with little saturation.

age GPT-4o to evaluate baselines that rely on fine-tuning on synthetic datasets, including HuatuoGPT-o1, R1-Distilled-Llama-8B and UltraMedical3.1-8B, where each model’s output is scored based on its medical reasonableness, logical coherence, and explainability. DrugDose is excluded from evaluation due to its small size and consequently unreliable statistical significance. The evaluation prompt is presented in Appendix E. Results in Fig. 3 indicates that MedS³ achieves the highest evaluation score. We attribute such superiority to the dual awareness of the PRM, which is trained to penalize wrong intermediate steps, and therefore could induce a relatively lower score to trajectories containing hallucinations and a correct final answer. The Best-of-N strategy avoids picking up such trajectories and enhances the interpretability of the final output.

5.3 Scaling of MedS³

In this section, we present the improvements brought by the self-evolutionary framework in Fig. 4a, and those attributable to test-time scaling in Fig. 4b. Specifically, we sample $n = 2, 4, 8, 16, 32, 64$ candidates for a prompt with a 1.0 temperature and compare the performance obtained through Best-of-N (BoN) (Lightman et al. 2023), PRM-guided Vote-Sum (P-VS; Wang et al. (2024)), as well as an SC baseline. We observe a great improvement in both the policy model and the PRM after a second evolution iteration, highlighting the efficacy of self-evolution. This suggests that the itera-

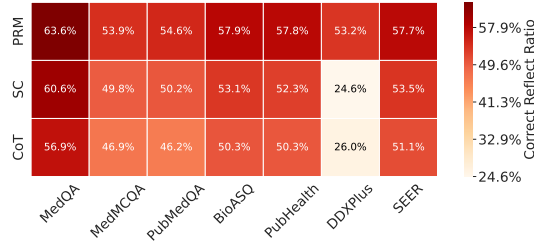


Figure 5: Reflective response ratio of MedS³ across 7 representative datasets. Both the policy and PRM are reflection-aware to perform sequential test-time scaling.

tive MCTS process, where the model learns from its own refined outputs, leads to steadily increased improvements. Additionally, we find that test-time scaling further enhances MedS³'s reasoning performance as illustrated in Fig. 4b in an effective log-linear rate with little saturation. Together, these results highlight the benefits of both self-exploration during synthesis and self-supervision during inference, contributing to MedS³'s strong performance across diverse tasks. Note that the P-VS performance is inferior to BoN, as most plausible reasoning chains arriving at correct answers but with incorrect reasoning steps are labeled with low values by our soft dual-sided PRM. Although these hallucinatory chains deteriorate the grouping of correct answers, our PRM still could assign the highest score for trajectories with both correct reasoning steps and final answers, therefore contributing to a log-linear scaling on BoN performance.

5.4 Introspective Behavior

Reflection has been proved to be an effective scaling paradigm for enhancing LLM's test-time scaling capacity (Guo et al. 2025). Our MedS³ introduced a Reflect node during synthesis and a soft dual-sided PRM to encourage correctly reflected responses, aiming to impart self-reflection behavior to the whole system. We manually define reflective tokens (*wait*, *reevaluate*, *recheck*, *however*, *but*) and count the ratio of correct responses with these tokens on seven representative benchmarks in Fig. 5. We observe a steady increase in the occurring ratio from directly chain-of-thought prompting to leveraging PRM to conduct BoN evaluation, which indicates both the policy and PRM in MedS³ has been imparted with self-reflection behavior. This further demonstrates that the PRM trained with the soft dual-sided label can correctly favor valuable responses with self-reflection.

5.5 Comparison of Reasoning Styles

In this section, we compare three reasoning enhancement strategies, including MCTS plus PRM which is what MedS³ leverages, with distillation from strong reasoning models, which is what O1-journey-part3 (Huang et al. 2025b) does and pure reinforcement learning (RL), which is what DeepSeek-R1 (Guo et al. 2025) adopts. We use the first iteration dataset in §3 to implement RL, and use the officially released distillation dataset provided by Huang et al. (2025b)

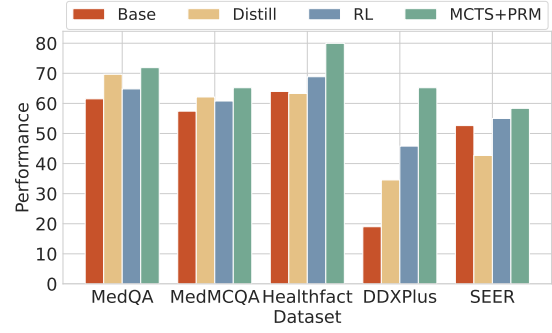


Figure 6: Three widely adopted methods to empower models with medical reasoning abilities. MCTS+PRM is the best among the three, making it the core of MedS³.

to SFT the base model, and compare them with MedS³ after the first evolution iteration. The results presented in Fig. 6 demonstrate that in exam-level medical QA datasets where the base model already excels at, distillation from large proprietary reasoning models is much more data-efficient than the other two methods, albeit sacrificing generalization in clinical tasks. In contrast, with both a considerable performance leap and generalization, RL is second to MCTS+PRM. We hypothesize that the soundness of medical diagnosis step is clear to determine, reducing reward hacking and resulting in a more reliable PRM and credible preference estimation.

6 Related Works

Slow-Thinking Medical LLMs With the significant achievements of the o1 (Jaech et al. 2024) in complex reasoning tasks, previous works show the potential advantage of the o1-like models in medical tasks (Xie et al. 2024; Xu et al. 2024; Nori et al. 2024). Based on these, previous works develop the slow-thinking medical LLMs with distillation: Huang et al. (2025b) directly learn the reasoning trajectory generated by o1 and Chen et al. (2024) improve the model's reasoning ability through o1 synthesis of reflective data and reinforcement learning. Besides, Yu et al. (2025) create a Chinese version slow-thinking medical LLMs by constructing the preference data with QwQ (Qwen 2024).

Self-Evolving Reasoning and Process Supervision Recent work in self-improving reasoning systems has explored Monte Carlo Tree Search (MCTS) (Zhang et al. 2024a) and reinforcement learning to enable models to refine their own outputs (Guo et al. 2025). Methods like Tree of Thoughts (ToT) (Yao et al. 2023) demonstrate the potential of search-based exploration for generating high-quality reasoning trajectories. Concurrently, process reward models (PRMs) have been proposed to provide step-wise feedback (Lightman et al. 2023), yet most assume binary correctness based on potential correctness and fail to penalize reasoning degradation.

7 Conclusion

In this paper, we present MedS³, a self-evolved slow-thinking system built for universal clinical usage. We extend the clini-

cal reasoning to diverse tasks to enhance generalization, and use MCTS to construct policy data and PRM data. We propose a new PRM learning objective – the soft dual-sided label, which enables the PRM to reward a step based on both future and past aspects, to produce credible long-chain reflective responses. Experiment results demonstrate that MedS³ achieves superior performance on diverse medical benchmarks, especially in realistic clinical ones, surpassing open-sourced models by a large margin with fewer parameters.

References

- Ankit Pal, M. S. 2024. OpenBioLLMs: Advancing Open-Source Large Language Models for Healthcare and Life Sciences. <https://huggingface.co/aaditya/OpenBioLLM-Llama3-70B>.
- Biswas, A.; and Talukdar, W. 2024. Intelligent Clinical Documentation: Harnessing Generative AI for Patient-Centric Clinical Note Generation. *arXiv preprint arXiv:2405.18346*.
- Cabral, S.; Restrepo, D.; Kanjee, Z.; Wilson, P.; Crowe, B.; Abdunour, R.-E.; and Rodman, A. 2024. Clinical reasoning of a generative artificial intelligence model compared with physicians. *JAMA internal medicine*, 184(5): 581–583.
- Chen, J.; Cai, Z.; Ji, K.; Wang, X.; Liu, W.; Wang, R.; Hou, J.; and Wang, B. 2024. HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs. *arXiv preprint arXiv:2412.18925*.
- Chen, Z.; Cano, A. H.; Romanou, A.; Bonnet, A.; Matoba, K.; Salvi, F.; Pagliardini, M.; Fan, S.; Köpf, A.; Mohtashami, A.; et al. 2023. Meditron-70b: Scaling medical pretraining for large language models. *arXiv preprint arXiv:2311.16079*.
- Christophe, C.; Kanithi, P. K.; Raha, T.; Khan, S.; and Pimentel, M. A. 2024. Med42-v2: A suite of clinical llms. *arXiv preprint arXiv:2408.06142*.
- DeYoung, J.; Lehman, E.; Nye, B.; Marshall, I.; and Wallace, B. C. 2020. Evidence Inference 2.0: More Data, Better Models. In Demner-Fushman, D.; Cohen, K. B.; Ananiadou, S.; and Tsujii, J., eds., *Proceedings of the 19th SIGBioMed Workshop on Biomedical Language Processing*, 123–132. Online: Association for Computational Linguistics.
- Dubey, A.; Jauhri, A.; Pandey, A.; Kadian, A.; Al-Dahle, A.; Letman, A.; Mathur, A.; Schelten, A.; Yang, A.; Fan, A.; Goyal, A.; Hartshorn, A.; Yang, A.; Mitra, A.; Sravankumar, A.; Korenev, A.; Hinsvark, A.; Rao, A.; Zhang, A.; Rodriguez, A.; Gregerson, A.; Spataru, A.; Rozière, B.; Biron, B.; Tang, B.; Chern, B.; Caucheteux, C.; Nayak, C.; Bi, C.; Marra, C.; McConnell, C.; Keller, C.; Touret, C.; Wu, C.; Wong, C.; Ferrer, C. C.; Nikolaidis, C.; Allonsius, D.; Song, D.; Pintz, D.; Livshits, D.; Esiobu, D.; Choudhary, D.; Mahajan, D.; Garcia-Olano, D.; Perino, D.; Hupkes, D.; Lakomkin, E.; AlBadawy, E.; Lobanova, E.; Dinan, E.; Smith, E. M.; Radenovic, F.; Zhang, F.; Synnaeve, G.; Lee, G.; Anderson, G. L.; Nail, G.; Mialon, G.; Pang, G.; Cucurell, G.; Nguyen, H.; Korevaar, H.; Xu, H.; Touvron, H.; Zarov, I.; Ibarra, I. A.; Kloumann, I. M.; Misra, I.; Evtimov, I.; Copet, J.; Lee, J.; Geffert, J.; Vranes, J.; Park, J.; Mahadeokar, J.; Shah, J.; van der Linde, J.; Billock, J.; Hong, J.; Lee, J.; Fu, J.; Chi, J.; Huang, J.; Liu, J.; Wang, J.; Yu, J.; Bitton, J.; Spisak, J.; Park, J.; Rocca, J.; Johnstun, J.; Saxe, J.; Jia, J.; Alwala, K. V.; Upasani, K.; Plawiak, K.; Li, K.; Heafield, K.; Stone, K.; and et al. 2024. The Llama 3 Herd of Models. *CoRR*, abs/2407.21783.
- Dubey, S.; Tiwari, G.; Singh, S.; Goldberg, S.; and Pinsky, E. 2023. Using machine learning for healthcare treatment planning. *Frontiers in Artificial Intelligence*, 6: 1124182.
- Guan, X.; Zhang, L. L.; Liu, Y.; Shang, N.; Sun, Y.; Zhu, Y.; Yang, F.; and Yang, M. 2025. rStar-Math: Small LLMs Can Master Math Reasoning with Self-Evolved Deep Thinking. *arXiv preprint arXiv:2501.04519*.
- Guo, D.; Yang, D.; Zhang, H.; Song, J.; Zhang, R.; Xu, R.; Zhu, Q.; Ma, S.; Wang, P.; Bi, X.; et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.
- Hendrycks, D.; Burns, C.; Basart, S.; Zou, A.; Mazeika, M.; Song, D.; and Steinhardt, J. 2021. Measuring Massive Multi-task Language Understanding. In *International Conference on Learning Representations*.
- Hu, E. J.; yelong shen; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*.
- Huang, X.; Wu, J.; Liu, H.; Tang, X.; and Zhou, Y. 2025a. m1: Unleash the Potential of Test-Time Scaling for Medical Reasoning in Large Language Models. *arXiv:2504.00869*.
- Huang, Z.; Geng, G.; Hua, S.; Huang, Z.; Zou, H.; Zhang, S.; Liu, P.; and Zhang, Z. 2025b. O1 Replication Journey – Part 3: Inference-time Scaling for Medical Reasoning. *arXiv preprint arXiv:2501.06458*.
- Huynh, T.; He, Y.; Willis, A.; and Rueger, S. 2016. Adverse Drug Reaction Classification With Deep Neural Networks. In Matsumoto, Y.; and Prasad, R., eds., *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, 877–887. Osaka, Japan: The COLING 2016 Organizing Committee.
- Jaech, A.; Kalai, A.; Lerer, A.; Richardson, A.; El-Kishky, A.; Low, A.; Helyar, A.; Madry, A.; Beutel, A.; Carney, A.; et al. 2024. Openai o1 system card. *arXiv preprint arXiv:2412.16720*.
- Jin, D.; Pan, E.; Oufattole, N.; Weng, W.-H.; Fang, H.; and Szolovits, P. 2021. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14): 6421.
- Jin, Q.; Dhingra, B.; Liu, Z.; Cohen, W.; and Lu, X. 2019. PubMedQA: A Dataset for Biomedical Research Question Answering. In Inui, K.; Jiang, J.; Ng, V.; and Wan, X., eds., *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2567–2577. Hong Kong, China: Association for Computational Linguistics.
- Jung, H.; Kim, Y.; Choi, H.; Seo, H.; Kim, M.; Han, J.; Kee, G.; Park, S.; Ko, S.; Kim, B.; et al. 2024. Enhancing Clinical Efficiency through LLM: Discharge Note Generation for Cardiac Patients. *arXiv preprint arXiv:2404.05144*.

- Khandekar, N.; Jin, Q.; Xiong, G.; Dunn, S.; Applebaum, S.; Anwar, Z.; Sarfo-Gyamfi, M.; Safranek, C.; Anwar, A.; Zhang, A.; et al. 2024. Medcalc-bench: Evaluating large language models for medical calculations. *Advances in Neural Information Processing Systems*, 37: 84730–84745.
- Kotonya, N.; and Toni, F. 2020. Explainable Automated Fact-Checking for Public Health Claims. In Webber, B.; Cohn, T.; He, Y.; and Liu, Y., eds., *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 7740–7754. Online: Association for Computational Linguistics.
- Kwon, W.; Li, Z.; Zhuang, S.; Sheng, Y.; Zheng, L.; Yu, C. H.; Gonzalez, J. E.; Zhang, H.; and Stoica, I. 2023. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles*.
- Lab, D. 2020. covid19-classification: Document Classification on COVID-19 Literature using the LitCovid collection and the Hedwig library. <https://github.com/dki-lab/covid19-classification>.
- Liao, Y.; Meng, Y.; Wang, Y.; Liu, H.; Wang, Y.; and Wang, Y. 2024. Automatic Interactive Evaluation for Large Language Models with State Aware Patient Simulator. *arXiv preprint arXiv:2403.08495*.
- Lightman, H.; Kosaraju, V.; Burda, Y.; Edwards, H.; Baker, B.; Lee, T.; Leike, J.; Schulman, J.; Sutskever, I.; and Cobbe, K. 2023. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*.
- McCreery, C. H.; Katariya, N.; Kannan, A.; Chablani, M.; and Amatriain, X. 2020. Effective transfer learning for identifying similar questions: matching user questions to COVID-19 FAQs. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining*, 3458–3465.
- Nori, H.; King, N.; McKinney, S. M.; Carignan, D.; and Horvitz, E. 2023. Capabilities of gpt-4 on medical challenge problems. *arXiv preprint arXiv:2303.13375*.
- Nori, H.; Usuyama, N.; King, N.; McKinney, S. M.; Fernandes, X.; Zhang, S.; and Horvitz, E. 2024. From Med-prompt to o1: Exploration of Run-Time Strategies for Medical Challenge Problems and Beyond. *arXiv preprint arXiv:2411.03590*.
- OpenAI. 2022. Chatgpt: Optimizing language models for dialogue. Website. <https://openai.com/blog/chatgpt>.
- OpenAI. 2023. GPT-4 Technical Report. *CoRR*, abs/2303.08774.
- Pal, A.; Umapathi, L. K.; and Sankarasubbu, M. 2022. Medmcqa: A large-scale multi-subject multi-choice dataset for medical domain question answering. In *Conference on health, inference, and learning*, 248–260. PMLR.
- Pappas, D.; Stavropoulos, P.; Androutsopoulos, I.; and McDonald, R. 2020. BioMRC: A Dataset for Biomedical Machine Reading Comprehension. In Demner-Fushman, D.; Cohen, K. B.; Ananiadou, S.; and Tsujii, J., eds., *Proceedings of the 19th SIGBioMed Workshop on Biomedical Language Processing*, 140–149. Online: Association for Computational Linguistics.
- Qwen. 2024. QwQ: Reflect Deeply on the Boundaries of the Unknown.
- Rafailov, R.; Sharma, A.; Mitchell, E.; Manning, C. D.; Ermon, S.; and Finn, C. 2023. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36: 53728–53741.
- Rajbhandari, S.; Rasley, J.; Ruwase, O.; and He, Y. 2020. Zero: Memory optimizations toward training trillion parameter models. In *SC20: International Conference for High Performance Computing, Networking, Storage and Analysis*, 1–16. IEEE.
- Shao, Z.; Wang, P.; Zhu, Q.; Xu, R.; Song, J.; Bi, X.; Zhang, H.; Zhang, M.; Li, Y.; Wu, Y.; et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.
- Singhal, K.; Azizi, S.; Tu, T.; Mahdavi, S. S.; Wei, J.; Chung, H. W.; Scales, N.; Tanwani, A.; Cole-Lewis, H.; Pfohl, S.; et al. 2023. Large language models encode clinical knowledge. *Nature*, 620(7972): 172–180.
- Tchango, A. F.; Goel, R.; Wen, Z.; Martel, J.; and Ghosn, J. 2022. DDXPlus Dataset.
- Tordjman, M.; Liu, Z.; Yuce, M.; Fauveau, V.; Mei, Y.; Hadjadj, J.; Bolger, I.; Almansour, H.; Horst, C.; Parihar, A. S.; et al. 2025. Comparative benchmarking of the DeepSeek large language model on medical tasks and clinical reasoning. *Nature medicine*, 1–1.
- Tsatsaronis, G.; Schroeder, M.; Paliouras, G.; Almirantis, Y.; Androutsopoulos, I.; Gaussier, E.; Gallinari, P.; Artieres, T.; Alvers, M. R.; Zschunke, M.; et al. 2012. Bioasq: A challenge on large-scale biomedical semantic indexing and question answering. In *2012 AAAI Fall Symposium Series*.
- Tu, T.; Palepu, A.; Schaeckermann, M.; Saab, K.; Freyberg, J.; Tanno, R.; Wang, A.; Li, B.; Amin, M.; Tomasev, N.; et al. 2024. Towards conversational diagnostic ai. *arXiv preprint arXiv:2401.05654*.
- Vilares, D.; and Gómez-Rodríguez, C. 2019. HEAD-QA: A Healthcare Dataset for Complex Reasoning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 960–966. Florence, Italy: Association for Computational Linguistics.
- Wang, P.; Li, L.; Shao, Z.; Xu, R.; Dai, D.; Li, Y.; Chen, D.; Wu, Y.; and Sui, Z. 2024. Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 9426–9439. Bangkok, Thailand: Association for Computational Linguistics.
- Wang, T.; Jiang, Z.; He, Z.; Yang, W.; Zheng, Y.; Li, Z.; He, Z.; Tong, S.; and Gong, H. 2025. Towards Hierarchical Multi-Step Reward Models for Enhanced Reasoning in Large Language Models. *arXiv preprint arXiv:2503.13551*.
- Wang, X.; Wei, J.; Schuurmans, D.; Le, Q. V.; Chi, E. H.; Narang, S.; Chowdhery, A.; and Zhou, D. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *The Eleventh International Conference on Learning Representations*.

- Wei, J.; Wang, X.; Schuurmans, D.; Bosma, M.; Xia, F.; Chi, E.; Le, Q. V.; Zhou, D.; et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35: 24824–24837.
- Welbl, J.; Liu, N. F.; and Gardner, M. 2017. Crowdsourcing Multiple Choice Science Questions. In Derczynski, L.; Xu, W.; Ritter, A.; and Baldwin, T., eds., *Proceedings of the 3rd Workshop on Noisy User-generated Text*, 94–106. Copenhagen, Denmark: Association for Computational Linguistics.
- Winands, M. H.; Björnsson, Y.; and Saito, J.-T. 2008. Monte-Carlo tree search solver. In *Computers and Games: 6th International Conference, CG 2008, Beijing, China, September 29–October 1, 2008. Proceedings 6*, 25–36. Springer.
- Wu, C.; Qiu, P.; Liu, J.; Gu, H.; Li, N.; Zhang, Y.; Wang, Y.; and Xie, W. 2024. Towards evaluating and building versatile large language models for medicine. *arXiv preprint arXiv:2408.12547*.
- Xie, Y.; Wu, J.; Tu, H.; Yang, S.; Zhao, B.; Zong, Y.; Jin, Q.; Xie, C.; and Zhou, Y. 2024. A Preliminary Study of o1 in Medicine: Are We Closer to an AI Doctor? *arXiv preprint arXiv:2409.15277*.
- Xu, S.; Zhou, Y.; Liu, Z.; Wu, Z.; Zhong, T.; Zhao, H.; Li, Y.; Jiang, H.; Pan, Y.; Chen, J.; Lu, J.; Zhang, W.; Zhang, T.; Zhang, L.; Zhu, D.; Li, X.; Liu, W.; Li, Q.; Sikora, A.; Zhai, X.; Xiang, Z.; and Liu, T. 2024. Towards Next-Generation Medical Agent: How o1 is Reshaping Decision-Making in Medical Scenarios. *CoRR*, abs/2411.14461.
- Xu, Z.; Jain, S.; and Kankanhalli, M. 2024. Hallucination is inevitable: An innate limitation of large language models. *arXiv preprint arXiv:2401.11817*.
- Yang, A.; Yang, B.; Hui, B.; Zheng, B.; Yu, B.; Zhou, C.; Li, C.; Li, C.; Liu, D.; Huang, F.; Dong, G.; Wei, H.; Lin, H.; Tang, J.; Wang, J.; Yang, J.; Tu, J.; Zhang, J.; Ma, J.; Xu, J.; Zhou, J.; Bai, J.; He, J.; Lin, J.; Dang, K.; Lu, K.; Chen, K.; Yang, K.; Li, M.; Xue, M.; Ni, N.; Zhang, P.; Wang, P.; Peng, R.; Men, R.; Gao, R.; Lin, R.; Wang, S.; Bai, S.; Tan, S.; Zhu, T.; Li, T.; Liu, T.; Ge, W.; Deng, X.; Zhou, X.; Ren, X.; Zhang, X.; Wei, X.; Ren, X.; Fan, Y.; Yao, Y.; Zhang, Y.; Wan, Y.; Chu, Y.; Liu, Y.; Cui, Z.; Zhang, Z.; and Fan, Z. 2024. Qwen2 Technical Report. *arXiv preprint arXiv:2407.10671*.
- Yao, S.; Yu, D.; Zhao, J.; Shafran, I.; Griffiths, T.; Cao, Y.; and Narasimhan, K. 2023. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36: 11809–11822.
- Yonsei. 2024. MedLLAMA3-v20. <https://huggingface.co/ProbeMedicalYonseiMAILab/medllama3-v20>.
- Yu, H.; Cheng, T.; Cheng, Y.; and Feng, R. 2025. FineMedLM-o1: Enhancing the Medical Reasoning Ability of LLM from Supervised Fine-Tuning to Test-Time Training. *arXiv preprint arXiv:2501.09213*.
- Zeng, Z.; Cheng, Q.; Yin, Z.; Zhou, Y.; and Qiu, X. 2025. Revisiting the Test-Time Scaling of O1-like Models: Do They Truly Possess Test-Time Scaling Capabilities? *arXiv:2502.12215*.
- Zhang, D.; Zhou, S.; Hu, Z.; Yue, Y.; Dong, Y.; and Tang, J. 2024a. ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search. *arXiv preprint arXiv:2406.03816*.
- Zhang, K.; Zeng, S.; Hua, E.; Ding, N.; Chen, Z.-R.; Ma, Z.; Li, H.; Cui, G.; Qi, B.; Zhu, X.; Lv, X.; Jinfang, H.; Liu, Z.; and Zhou, B. 2024b. UltraMedical: Building Specialized Generalists in Biomedicine. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Zhang, Z.; Zheng, C.; Wu, Y.; Zhang, B.; Lin, R.; Yu, B.; Liu, D.; Zhou, J.; and Lin, J. 2025. The Lessons of Developing Process Reward Models in Mathematical Reasoning. *arXiv preprint arXiv:2501.07301*.
- Zhao, Z.; Jin, Q.; Chen, F.; Peng, T.; and Yu, S. 2023. A large-scale dataset of patient summaries for retrieval-based clinical decision support systems. *Scientific Data*, 10.
- Zotova, E.; Agerri, R.; Nuñez, M.; and Rigau, G. 2020. Multilingual Stance Detection in Tweets: The Catalonia Independence Corpus. In Calzolari, N.; Béchet, F.; Blache, P.; Choukri, K.; Cieri, C.; Declerck, T.; Goggi, S.; Isahara, H.; Maegaard, B.; Mariani, J.; Mazo, H.; Moreno, A.; Odijk, J.; and Piperidis, S., eds., *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 1368–1375. Marseille, France: European Language Resources Association. ISBN 979-10-95546-34-4.
- Zuo, Y.; Qu, S.; Li, Y.; Chen, Z.; Zhu, X.; Hua, E.; Zhang, K.; Ding, N.; and Zhou, B. 2025. Medxpertqa: Benchmarking expert-level medical reasoning and understanding. *arXiv preprint arXiv:2501.18362*.

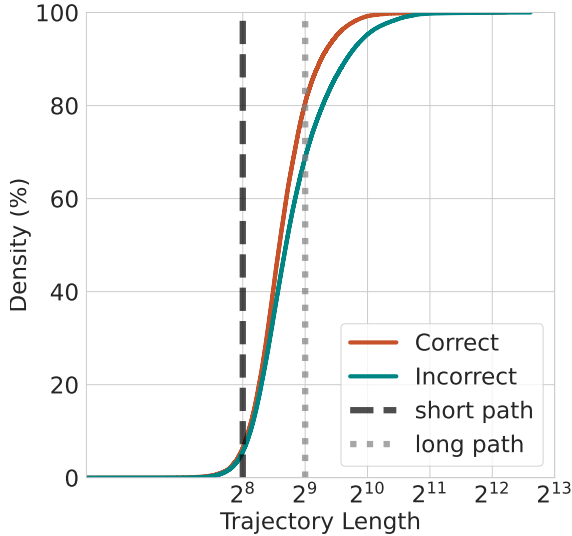


Figure 7: Trajectory length distribution of correct and incorrect sets of the evolved dataset.

A Statistics of the Evolved Dataset

In this section, we show the statistics of the evolved dataset after the second evolution, which is used to conduct our training of MedS³. We plot the length distributions of trajectories in Fig. 7. Defining short sequences as below 256 and long sequences as above 512, we find that the evolved dataset contains about 20% long trajectories, which enables the policy model to generate reliable responses with more tokens. Moreover, we find that correct trajectories consume fewer tokens than incorrect ones, which aligns highly with Zeng et al. (2025).

B Limitations

MedS³ achieves superior performance over eleven benchmarks by conducting MCTS in seed datasets to collect both policy and PRM training data and a newly proposed PRM learning objective: soft dual side label. However, it can be further improved via these strategies: (1) adopt human-in-the-loop strategies to further enhance the interpretability of medical reasoning LLMs; (2) introduce more training samples to cover more medical reasoning scenarios; (3) conduct more evolution iterations to further improve the model.

C Further Experiments

In this section, we present more experiments to validate the effectiveness of MedS³.

C.1 Determination of β

We perform a simple grid search on a pre-defined dev set to find the most appropriate value of β in Eq. 7. Specifically,

β	0.5	1	1.5	2
Dev loss	0.4293	0.4169	0.4194	0.538

Table 5: Grid search of β and corresponding loss in the dev set.

we search β in the following list [0.5, 1.0, 1.5, 2.0] and show the loss in the development set in Table 5. We determine β as 1 for its lowest loss. Although there might exist a more advanced configuration, we just set β to 1 as this is not our focus and we leave this for future work.

C.2 Comparison with SC Models

We also compare MedS³ with baselines prompted with the Self-Consistency (SC) method, which is a simple yet efficient way to scale in a parallel manner. To maintain similar inference costs, we set the sampling number to 32 for models with similar size (<10B) and compare with the most powerful models before², namely HuatuoGPT-o1 and UltraMedical3.1-8B. Results in Table 6 illustrate that as a test-time scaling method, SC improves the already strong baselines by significant gains, while such improvements usually occur in traditional benchmarks like MedQA or MedMCQA. Their performances in clinical testbeds, like SEER or DDX-Plus, hardly show gains, which unveils some kind of overfitting problem. Therefore, optimized for both traditional examination and clinical usage, MedS³ robustly achieves state-of-the-art performance overall.

C.3 Backbone Selection

In this section, we investigate which backbone, a general LLM or a medical-specific LLM, is suitable for conducting self-evolution. We conduct one iteration of evolution using the same data of MedS³ under UltraMedical3.1-8B and compare it with MedS³ after the **first iteration** using the Best-of-N decoding method to save computational cost. The comparison shown in the upper half of Table 7 reveals that although UltraMedical3.1-8B improves compared to the initial policy, it still lags behind MedS³ by a large margin. Delving into the generation, we find that UltraMedical3.1-8B suffers from endless generation, which stems from its lower instruction following ability compared to Llama 3.1 8B. On the other hand, medical backbones show no significant performance gains compared to the general model (UltraMedical3.1 8B 60.11 vs Llama 3.1 8B 58.98 in Table 2), while after optimized for certain benchmarks like MedQA, they have lower generalization ability than general models. Based on the above observations, we choose to use a general backbone with sufficient medical knowledge, i.e., Llama 3.1 8B, as the initial policy model.

D Future Work

As a pioneering work, we have validated that small language models can self-evolve to empower themselves with strong

²It is reasonable when MedS³ outperforms the most leading baselines.

Model	MedQA	MedMCQA	PubmedQA	Bioasq	Med MMLU	Biomrc	Pubhealth	Healthfact	DDX Plus	Drug Dose	SEER	Average
Ultramedical3.1-8B	72.90	65.57	58.00	80.58	78.54	49.18	67.91	71.72	34.10	60.87	49.73	62.65
m1-7b-32K	72.38	64.55	50.40	80.91	80.19	59.50	58.08	68.10	31.80	82.61	50.91	63.58
HuatuoGPT-o1	72.86	66.94	58.20	78.48	78.54	46.45	64.58	70.29	41.00	60.87	49.59	62.81
MedS ³	72.97	67.32	64.20	81.39	79.63	74.54	74.41	76.18	62.40	91.30	59.80	73.10

Table 6: Comparison with prior Top-2 models with similar model sizes prompted with self-consistency method. Albeit certain improvements, these models still lag behind MedS³ by a large margin.

Backbone	MedQA	MedMCQA	PubMedQA	BioASQ	Med MMLU	BioMRC	PubHealth	HealthFact	DDX Plus	Drug Dose	SEER	Average
UltraMedical3.1	68.42	58.20	58.00	79.61	73.16	49.40	68.07	71.38	49.20	86.96	51.40	64.89
Llama 3.1	67.64	62.00	59.60	79.13	77.77	76.96	73.19	78.37	63.80	91.30	59.20	71.72

Table 7: Comparison with UltraMedical-3.1-8B as the policy model. With superior instruction following ability and comparable medical knowledge, Llama-3.1-8B suits MedS³ system to fulfill the self-evolution procedure.

Method	Iteration	MedQA	MedMCQA	PubmedQA	Bioasq	Med MMLU	Biomrc	Pubhealth	Healthfact	DDX Plus	Drug Dose	SEER	Average
BoN	2	68.97	64.04	62.00	79.45	76.43	73.68	72.14	73.57	58.00	86.96	55.45	70.06
	4	69.60	64.55	61.60	80.42	77.26	74.64	74.17	73.06	58.85	86.96	56.75	70.71
	8	70.54	64.57	62.60	81.07	77.83	74.75	74.09	72.64	59.90	86.96	56.61	71.05
	16	70.23	66.32	64.00	81.23	78.41	74.80	73.68	72.05	61.00	86.96	58.44	71.56
	32	72.97	67.32	64.20	81.39	79.63	74.54	74.41	76.18	62.40	91.30	59.80	73.10
	64	73.37	67.65	66.00	81.72	79.37	74.54	74.90	78.28	62.25	91.30	60.79	73.65
SC	2	65.67	61.49	60.60	77.02	73.73	71.09	68.48	70.79	56.45	91.30	52.59	68.11
	4	67.09	63.11	60.40	78.80	75.72	73.23	70.59	76.18	57.35	91.30	56.32	70.01
	8	67.40	63.71	60.60	80.42	76.30	73.82	70.11	77.61	57.65	91.30	57.48	70.58
	16	68.42	63.73	60.80	80.42	76.43	73.70	70.11	77.69	58.05	91.30	58.27	70.81
	32	67.64	63.52	60.60	80.26	76.55	73.98	70.59	78.28	57.90	91.30	58.25	70.81
	64	67.79	63.45	60.80	80.26	76.75	73.98	70.76	77.86	58.10	91.30	58.33	70.85
P-VS	2	68.97	64.04	62.00	79.45	76.43	73.68	72.14	73.57	58.00	86.96	55.45	70.06
	4	68.97	63.85	60.40	80.26	75.91	74.38	71.57	75.25	57.70	86.96	57.37	70.24
	8	68.34	63.95	61.00	80.74	76.55	74.54	70.11	76.60	55.40	86.96	57.37	70.14
	16	68.81	63.88	60.80	81.39	77.07	74.99	70.27	75.67	53.10	86.96	58.33	70.12
	32	68.66	63.81	61.20	80.74	76.81	74.88	71.16	74.41	53.65	82.61	57.99	69.63
	64	68.19	63.71	61.00	80.58	77.39	74.88	71.24	74.41	53.70	82.61	58.16	69.62

Table 8: Full table of test-time scaling using PRM with different evaluation methods.

reasoning abilities in clinical usage. There are several remaining directions to further enhance MedS³:

1. **Conduct Human-interference evaluation.** MC-rollout value is verified to be not the best choice for evaluating the value of an internal step. We are eager to introduce a more fine-grained step label to enhance the optimization of the PRM.
2. **Introduce more clinical data, not limited to close-ended generation.** Currently, all the data used in MedS³ are close-ended, and the application of reasoning is not limited to such a narrow room. We intend to extend MedS³ to broader clinical tasks to make MedS³ a more useful system.

We will continue our exploration and make MedS³ more practical in medical domains.

E Prompt Template

We show the prompt used to synthesize reasoning data in Fig. 8, Fig. 9, and Fig. 10. The prompt template to evaluate the hallucination of each medical LLM using synthetic data is shown in Fig 11.

F Dataset Details

In this section, we elucidate the seed dataset and the evaluation sets. We also clearly denote the involved dataset’s usage during training and evaluation and their corresponding category in Table 9. We divide the used 16 training datasets into the following five dimensions:

1. **Long Context QA:** This dimension enables MedS³ to capture useful information from the given context and response with long-chain reasoning. This dimension covers BioMRC (Pappas et al. 2020), HeadQA Topic Classification (Vilares and Gómez-Rodríguez 2019; Wu et al. 2024), and HealthFact (Kotonya and Toni 2020)
2. **Knowledge-Intensive QA:** This dimension teaches MedS³ to use long-chain reasoning to answer knowledge-intensive problems, which covers MedQA (Jin et al. 2021), MedMCQA (Pal, Umapathi, and Sankarasubbu 2022), and PubMedQA (Jin et al. 2019).
3. **Bio-Medical QA:** This part leverages general data in biomedicine domains to enhance the generality of MedS³, which includes SciQ (Welbl, Liu, and Gardner 2017), Evidence Inference (DeYoung et al. 2020) and Head QA (Vilares and Gómez-Rodríguez 2019).
4. **Medical Natural Language Inference:** This dimension prompts MedS³ to discriminate biomedical research concepts and corresponding descriptions, which contain PubHealth (Kotonya and Toni 2020), Medical Question Pair (MQP; McCreery et al. (2020)), and catalonia-independence-corpus (CIC; Zotova et al. (2020)).
5. **Diagnosis QA:** This dimension is related to real-world clinical scenarios, including disease diagnosis and classification and drug related questions. We choose Covid-19 Classification (Lab 2020), Drug-Dose Extraction, Adverse Drug Event Classification (Huynh et al. 2016; Wu et al. 2024) and DDX-Plus (Tchango et al. 2022)..

The descriptions of each training and evaluation dataset are presented below:

1. **MedQA** (Jin et al. 2021) is a widely used benchmark for evaluating AI systems in medical question answering, featuring multiple-choice questions from professional medical licensing exams such as the USMLE and exams from China and Taiwan. We adopt its 5-options English version, taking its training set as seed data and 1,273 test problems as the evaluation benchmark.
2. **PubmedQA** (Jin et al. 2019) is a specialized benchmark for biomedical question answering, consisting of question-answer pairs derived from PubMed abstracts. It focuses on yes/no/maybe questions that require reasoning over biomedical literature. We use the human-labeled question set and split the training set and test set, with both 500 problems for evolution and evaluation, respectively. Note that we do not include relevant contexts before questions, challenging models’ internal knowledge comprehension.
3. **MedMCQA** (Pal, Umapathi, and Sankarasubbu 2022) is a large-scale benchmark for medical question answering, featuring over 194,000 multiple-choice questions sourced from Indian medical entrance exams and other educational resources. It spans a wide range of medical topics, including anatomy, pharmacology, and pathology, and is designed to evaluate the reasoning and knowledge application skills of AI systems in a clinical context. The test set contains 4,183 problems.
4. **MMLU** (Hendrycks et al. 2021) is to measure LLM’s multitask accuracy, which contains 14,421 problems. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more. We select its medical-related problems, resulting in a test set with 1,561 problems.
5. **BioMRC** (Pappas et al. 2020) is a collection of medical-related question-answer pairs, specifically designed for the evaluation of machine reading comprehension (MRC) tasks in the biomedical domain. It is derived from a wide range of medical texts, including clinical notes, research papers, and medical textbooks. The dataset contains a series of questions and corresponding answers, where the answers are extracted from relevant passages of text. We use its 6,250 test set as the evaluation set.
6. **HeadQA** (Vilares and Gómez-Rodríguez 2019) is a specialized medical question-answering dataset designed to evaluate models in the context of neurology and head-related disorders. It consists of a collection of questions paired with answers derived from a variety of clinical notes, medical reports, and other head-related health data sources.
7. **DDX-Plus** (Tchango et al. 2022) is a comprehensive medical diagnostic dataset designed to assist in the development and evaluation of machine learning models for differential diagnosis in clinical settings. It consists of clinical cases, where each case includes a set of symptoms, patient history, physical examination findings, and diagnostic questions, along with a list of potential diagnoses ranked by their likelihood. The diverse set of cases

Reason Template

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023 Today Date: 23 July 2024  
  
<|eot_id|><|start_header_id|>user<|end_header_id|>  
  
Reasoning Example: {Few-shot Example}  
  
You are a professional medical expert majored at reasoning in hard  
medical-related problems.  
  
Think critically about the problem and answer with concise, accurate reasoning.  
Please ensure your reasoning is thorough and elaborate, breaking down each step  
of your thought process.  
  
Problem: {problem}<|eot_id|><|start_header_id|>assistant<|end_header_id|>  
  
Step 0: Let's break down this problem step by step  
  
Step 1:
```

Figure 8: Reason template

Finish Template

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023 Today Date: 23 July 2024  
  
<|eot_id|><|start_header_id|>user<|end_header_id|>  
  
Reasoning Example: {Few-shot Example}  
  
You are a professional medical expert majored at reasoning in hard  
medical-related problems.  
  
Use thorough and elaborate steps to complete your reasoning. Conclude the task  
by stating: "The answer is {answer}".  
  
Problem: {problem}<|eot_id|><|start_header_id|>assistant<|end_header_id|>  
  
Step 0: Let's break down this problem step by step  
  
Step 1:
```

Figure 9: Finish template

Reflect Template

<|begin_of_text|><|start_header_id|>system<|end_header_id|>

Cutting Knowledge Date: December 2023 Today Date: 23 July 2024

<|eot_id|><|start_header_id|>user<|end_header_id|>

Reasoning Example: {Few-shot Example}

You are a professional medical expert majored at reasoning in hard medical-related problems.

Use thorough and elaborate steps to complete your reasoning. Conclude the task by stating: "The answer is {answer}".

Problem: {problem}<|eot_id|><|start_header_id|>assistant<|end_header_id|>

Step 0: Let's break down this problem step by step

Step 1: [omitted]

Step k: [omitted]. The answer is C.

Step k+1: Wait, the previous answer maybe incorrect and I need to reconsider other options.

Figure 10: Reflect template

Evaluate Template

Given a medical problem and a response from a large language model (whose final prediction is correct), please based on the following criteria to give a score:

Reasonableness: The response should be reasonable and consistent with the medical problem.

Coherence: The response should be coherent and logically consistent.

Explainability: The response should be explainable and easy to understand.

Please give a score from 0 to 10, where 0 means the response is completely unreasonable, and 10 means the response is perfect. Please also provide a brief explanation of your score.

Give your score in the following format:

<score>{{Your score}}</score>

Question: {question}

Response: {response}

Figure 11: GPT Evaluation template

in the dataset spans multiple medical specialties, making it an ideal resource for creating models capable of assisting healthcare professionals in making informed diagnostic decisions. Due to its huge test set (over 100,000 test instances), we randomly select 2,000 items for evaluation.

8. SciQ (Welbl, Liu, and Gardner 2017) is a scientific question-answering dataset designed to assess machine learning models in answering factual questions across a wide range of scientific domains. It consists of over 13,000 questions derived from scientific literature, including topics in physics, biology, chemistry, and earth sciences, among others. Each question is paired with a correct answer and is supported by a passage of text from which the answer is extracted.
9. Evidence Inference (DeYoung et al. 2020) is a collection designed to evaluate machine learning models on their ability to infer logical conclusions from evidence presented in the form of textual information. This dataset consists of structured pairs of premises (evidence) and hypotheses, where the goal is for models to determine the logical relationship between them—whether the hypothesis is supported, contradicted, or is neutral with respect to the provided evidence. Typically used for tasks such as textual entailment or natural language inference (NLI), the dataset includes a variety of complex scenarios across multiple domains, including law, healthcare, and science, where reasoning based on available evidence is crucial.
10. PubHealth (Kotonya and Toni 2020) is a comprehensive dataset for explainable automated fact-checking of public health claims. Each instance in the PUBHEALTH dataset has an associated veracity label (true, false, unproven, mixture). Furthermore, each instance in the dataset has an explanation text field. The explanation is a justification for which the claim has been assigned a particular veracity label. We construct two different test sets. Healthfact is to directly predict whether a given instance is true/false/unproven/mixture. The other, Pubhealth, is to predict whether the instance sentence and the given explanation express the same meaning.
11. Medical Question Pair (McCreery et al. 2020) contains a dataset of 3,048 similar and dissimilar medical question pairs hand-generated and labeled by Curai’s doctors. Models should clarify whether the two given questions are similar or not.
12. Catalonia-independence-Corpus (Zotova et al. 2020) is a dataset built for stance detection in Twitter for the Catalan and Spanish languages, with the aim of facilitating research on stance detection in multilingual and cross-lingual settings.
13. Covid-19 Classification (Lab 2020) is an extension of the Hedwig library and contains all necessary code to reproduce the results of some document classification models on a COVID-19 dataset created from the LitCovid collection.
14. Adverse Drug Event (Huynh et al. 2016) is critical for developing automated systems that can support clinicians in identifying harmful drug reactions, potentially reducing healthcare costs, and enhancing patient safety. Given

Category	Dataset	Train	Test
Diagnosis QA	ADE	Yes	No
	Covid-19 CLS	Yes	No
	DrugDose	Yes	Yes
	DDXPlus	Yes	Yes
	SEER	No	Yes
Medical NLI	PubHealth	Yes	Yes
	CIC	Yes	No
	MQP	Yes	No
Long Context QA	BioMRC	Yes	Yes
	HealthFact	Yes	Yes
	HeadQA Topic CLS	Yes	No
BioMedical QA	HeadQA	Yes	No
	Evidence Extraction	Yes	No
	SciQ	Yes	No
Knowledge QA	MedQA	Yes	Yes
	MedMCQA	Yes	Yes
	PubMedQA	Yes	Yes
	MMLU	No	Yes
	BioASQ	No	Yes

Table 9: Medical datasets usage during training and evaluation. “CLS” denotes classification.

the increasing volume of clinical data, this dataset plays a key role in advancing AI-driven drug safety research and improving the overall quality of healthcare. We build Drugdose extraction test set to benchmark models to extract the exact dose of a specific drug.

15. SEER (Dubey et al. 2023) is purposed for treatment planning because it contains key clinical variables that directly inform therapy decisions (e.g., tumor size, nodal status, hormone receptor status). LLMs must answer the most appropriate suggestion from the following list [‘Intraoperative rad with other rad before/after surgery’, ‘Intraoperative radiation’, ‘No radiation and/or cancer-directed surgery’, ‘Radiation after surgery’, ‘Radiation before and after surgery’, ‘Radiation prior to surgery’, ‘Surgery both before and after radiation’] based on patient summarization, simulating real-world tumor board decisions.

G Hyperparameters

G.1 Data Synthesis

For each node expansion, we simultaneously generate 3 different responses with the same prompt. We set the generation temperature to 1. The stop tokens are set to {Step k: | $k = 1, 2, \dots, 100$ } to ensure that each node represents a single reasoning step. We use the first sample in MedQA as the one-shot example and prompt GPT-4o to generate step-by-step outputs.

G.2 Self-Training of Policy and PRM

We use 8xNVIDIA A100 GPUs and the overall training consumes 14h.

Policy tuning We use trl³ as the training framework. We first use vanilla SFTTrainer to train the policy model. We set the warmup ratio to 0.03 and the max sequence length to 8192. The batch size is set to 128 and the learning rate is set to 1e-6. After that, we use DPOTrainer to further fine-tune the policy model. We set the learning rate to 5e-8 and the batch size to 128.

PRM tuning We use PRMTrainer of trl to train the PRM model. We use LoRA to fine-tune the PRM, where the lora rank is set to 32 and lora alpha set to 64. The learning rate is set to 5e-5. For a single step s_k , the input for PRM is the concatenation of all steps up to the current step, namely:

$$P = s_0 \oplus s_1 \oplus \dots \oplus s_k \quad (9)$$

$$\hat{y} = V_\theta(P; x) \quad (10)$$

This input models a step’s value with causal relationships between steps, preventing local optima learning.

G.3 Evaluation

For evaluation, the temperature is set to 1.0 and top_p is set to 0.9 for multiple sampling settings. For the comparison between policy models, the temperature is set to 0 and the greedy decoding method is adopted to avoid variable results. The max generation tokens are set to 8,192. Among the three presented decoding mechanisms, CoT (Wei et al. 2022) directly prompts models to generate a long reasoning chain and outputs the answer with “The answer is {answer}” for the convenience of answer extraction. Self-Consistency (Wang et al. 2023) generates $N = 32$ samples for a given problem, and we select the one whose answer appears most times among the N outputs. We use exact match (EM) to measure the performance. Specifically, we extract the contents following the last “The answer is” template to match the self-reflection thinking style, and perform appropriate post-processing to derive a final prediction. No matter what the contents the model has generated, we manually append “The answer is” to the end of the generation, and prompt the model to continue generation, with a small token limit (20) and pre-set logit bias to ensure the generation falls into candidate tokens. For multiple-choice problems, we directly choose the first character of prediction phrases and measure whether the ground truth is equal to the prediction. For close-ended generation tasks, we remove quotes and turn the prediction and the ground truth into lowercase phrases. After that, we check whether the ground truth phrases exist in the prediction phrases.

G.4 Training Details of Distillation and RL

In this section, we elucidate the implementation details of distillation and RL.

Distillation For Distillation method, we fine-tune Llama3.1-8B with 2K training data⁴ released by Huang et al. (2025b), which combined with the questions in MedQA

and corresponding response generated by o1 (Jaech et al. 2024). We adopt LoRA (Hu et al. 2022) and set the rank r to 16 and alpha α to 32 for fair comparisons. For fine-tuning parameters, we set the learning rate to 2e-6 and batch size to 128.

RL We follow Guo et al. (2025) to use Group Relative Policy Optimization (GRPO; Shao et al. (2024)) to conduct RL training. We set the number of generations to 10 and the learning rate to 1e-6. We adopt ZeRO-3 (Rajbhandari et al. 2020) to save memory and conduct full fine-tuning in one 8xA100 machine. The batch size is set to 4 per GPU. For the adopted prompt, we use <think></think><answer></answer> to learn the slow-thinking output style. We use accuracy reward and format reward, where

$$r = \begin{cases} 1 & \text{Correct answer with correct format} \\ 0 & \text{Incorrect answer with correct format} \\ -1 & \text{Incorrect format} \end{cases} \quad (11)$$

H Best-of-N Details

In this section, we elucidate the fast inference using Best-of-N (BoN) evaluation with the PRM. Specifically, the policy model generates N responses $\{y_i \mid i \in [1, N]\}$ simultaneously using the inference engine (vLLM; Kwon et al. (2023)). After that we split each response y_i into K steps: $[s_i^1, s_i^2, \dots, s_i^K]$ where $k \in [1, K]$ using predefined step separator `Step`. We form the step input to the PRM (x, S_i^k) as $S_i^k = s_i^1 \oplus s_i^2 \oplus \dots \oplus s_i^k$ and obtain each step’s value $v_i^k = \text{softmax}(V_\theta(x; S_i^k))$ using a forward pass. The value is the probability of predicting the step to be correct, which is in the range $[0, 1]$. Finally, the whole response value $v_i = \min_{k=1}^K v_i^k$ is taken from the minimum value of each step and we select the response with the highest value.

I Case Study

³<https://huggingface.co/docs/trl/index>

⁴<https://huggingface.co/datasets/SPIRAL-MED/o1-journey-Ophiuchus>

Question:

A 47-year-old executive schedules an appointment his physician for a routine medical check-up. He currently has no complaints and claims to be "as fit as a fiddle." The physical examination findings are unremarkable, except for a mid-systolic murmur heard in the 2nd left intercostal space that radiates to the carotids on auscultation. The physician instructs the patient to stand from a supine position with the stethoscope still placed on his chest. Which of the following changes would occur with this maneuver?\nA. An increase in right atrial pressure\nB. An increase in left ventricular end-diastolic pressure\nC. An increase in pulmonary capillary wedge pressure\nD. A reduction in the slope of the pacemaker potential\nE. A reduction in diastolic filling time

Llama3.1-8B:

When the patient stands from a supine position, the following changes occur:\n\n* Venous return to the heart decreases due to gravity, which leads to a decrease in right atrial pressure (A).\n* The decrease in venous return also leads to a decrease in left ventricular end-diastolic pressure (B).\n* The decrease in left ventricular end-diastolic pressure would not directly affect the pulmonary capillary wedge pressure (C).\n* The pacemaker potential is not directly affected by changes in posture.\n* The diastolic filling time is not directly affected by changes in posture.\n\nThe mid-systolic murmur heard in the 2nd left intercostal space that radiates to the carotids is likely due to aortic stenosis. **When the patient stands from a supine position, the murmur may become louder due to the increased gradient across the aortic valve.** \n\nThe answer is A.

MedS³

Step 0: Let's break down this problem step by step.\n\nStep 1: Identify the likely diagnosis based on the physical examination findings. The presence of a mid-systolic murmur heard in the 2nd left intercostal space that radiates to the carotids on auscultation is suggestive of aortic stenosis. Aortic stenosis is characterized by the narrowing of the aortic valve opening, which can lead to a delay in left ventricular ejection and a subsequent murmur.\n\nStep 2: Understand the effect of standing from a supine position on the cardiovascular system. When a person stands from a supine position, gravity causes a decrease in venous return to the heart. This decrease in venous return leads to a decrease in the preload, which is the initial stretching of the cardiac myocytes prior to contraction.\n\nStep 3: Determine how the decrease in preload affects the heart's function. A decrease in preload leads to a decrease in the end-diastolic volume, which in turn leads to a decrease in the stroke volume. This is because the heart's ability to pump blood is directly related to the amount of blood it receives during diastole.\n\nStep 4: Consider how the decrease in stroke volume affects the murmur associated with aortic stenosis. The murmur associated with aortic stenosis is typically a systolic ejection murmur, which is caused by the delay in left ventricular ejection. When the stroke volume decreases, the murmur may become softer or less intense, but it will not change its timing or location.\n\nStep 5: Analyze the options provided to determine which one is most closely related to the change in the murmur associated with aortic stenosis. The murmur is not directly related to the options provided, but the decrease in stroke volume is related to the decrease in diastolic filling time. When the heart is in a supine position, the diastolic filling time is longer due to the increased venous return. When the person stands, the diastolic filling time decreases due to the decreased venous return.\n\nStep 6: Select the correct answer based on the analysis. The correct answer is the one that is most closely related to the change in the murmur associated with aortic stenosis, which is the decrease in diastolic filling time.\n\n**The answer is E.**

MedS³-PRM:

Step 0: Let's break down this problem step by step.\n\nStep 1: Identify the initial clinical scenario and physical examination findings. The patient is an asymptomatic 47-year-old executive undergoing a routine check-up. The physical examination reveals a mid-systolic murmur heard at the 2nd left intercostal space radiating to the carotids. This murmur is suggestive of a pulmonary valve issue, but given the radiation to the carotids, it's more likely to be a semilunar valve issue, and the closest one would be the aortic valve, however the classic exam for aortic stenosis is the chest pressure, and the one described would be more indicative of a pulmonary valve issue but in the given location it's likely aortic stenosis. However, the key piece of information here is the physician's instruction to have the patient stand from a supine position while continuing to auscultate the heart.\n\nStep 2: Understand the effect of changing body position on cardiovascular physiology. When a person transitions from a supine (lying down) position to a standing position, several physiological changes occur:\n\n- Venous blood pools in the lower extremities due to gravity, reducing venous return to the heart.\n- Cardiac output decreases due to reduced preload.\n- Heart rate may increase to maintain cardiac output (baroreceptor reflex).\n- Systemic vascular resistance may increase to compensate for the decreased cardiac output.\n\nThe specific maneuver instructed by the physician, where the patient stands from a supine position while continuing to auscultate the heart, will lead to a decrease in venous return and, consequently, a decrease in left ventricular preload. This change in preload can affect the intensity of heart sounds, particularly those related to ventricular filling.\n\nStep 3: Relate the physiological changes to the potential effects on the heart sounds and the murmur in question. In the case of a mid-systolic murmur, as described, the intensity of the murmur is more likely related to the flow across the valve during systole. However, in this scenario, the key is to understand how the change in body position affects the physiology of the heart and the likelihood of certain effects on heart sounds and pressures.\n\nGiven the change in position, the most likely effect on the pressures and heart sounds would be a decrease in the diastolic filling time due to decreased venous return and thus a decrease in left ventricular end-diastolic pressure and volume, however the best answer given would be a decrease in diastolic filling time due to decreased preload, although the best answer regarding a direct relationship between the two isn't given in the question options.\n\nHowever, we must find the closest relation. If we consider the decrease in preload, and looking at the options:\n\n- A. An increase in right atrial pressure is not directly related to the change in body position as described.\n- B. An increase in left ventricular end-diastolic pressure is the opposite of what would be expected with decreased preload.\n- C. An increase in pulmonary capillary wedge pressure is also not directly related to the change in body position.\n- D. A reduction in the slope of the pacemaker potential is related to the effect of decreased preload on the sinoatrial node but is more related to the baroreceptor reflex rather than the direct effect of decreased preload on the ventricles.\n- E. A reduction in diastolic filling time is the closest option, as it is directly related to the decrease in preload due to the change in body position.\n\n**The answer is E.**

Figure 12: Cases of MedQA

Question:

Imagine you are a doctor, next I will give you a dialogue and please make a diagnosis decision based on it. You only need to choose one answer among the following lists: ['spontaneous pneumothorax', 'cluster headache', 'boerhaave', 'spontaneous rib fracture', 'gerd', 'hiv (initial infection)', 'anemia', 'viral pharyngitis', 'inguinal, hernia', 'myasthenia gravis', 'whooping cough', 'anaphylaxis', 'epiglottitis', 'guillain-barré syndrome', 'acute laryngitis', 'croup', 'psvt', 'atrial fibrillation', 'bronchiectasis', 'allergic sinusitis', 'chagas', 'scombroid food poisoning', 'myocarditis', 'larygospasm', 'acute dystonic reactions', 'localized edema', 'sle', 'tuberculosis', 'unstable angina', 'stable angina', 'ebola', 'acute otitis media', 'panic attack', 'bronchospasm / acute asthma exacerbation', 'bronchitis', 'acute copd exacerbation / infection', 'pulmonary embolism', 'urti', 'influenza', 'pneumonia', 'acute rhinosinusitis', 'chronic rhinosinusitis', 'bronchiolitis', 'pulmonary neoplasm', 'possible nstemi / stemi', 'sarcoidosis', 'pancreatic neoplasm', 'acute pulmonary edema', 'pericarditis', 'cannot decide']. The answer should be a single word. \nAge: 45 \n\nSex: M \n\nInitial evidence: Do you have a cough that produces colored or more abundant sputum than usual? Yes \n\nEvidence: Do you have pain somewhere, related to your reason for consulting? Yes; Characterize your pain: Sensitive; Do you feel pain somewhere? Lower chest; Do you feel pain somewhere? Side of the chest(r); Do you feel pain somewhere? Breast(r); Do you feel pain somewhere? Posterior chest wall(r); Do you feel pain somewhere? Posterior chest wall(l); How intense is the pain? 2; Does the pain radiate to another location? Nowhere; How precisely is the pain located? 5; How fast did the pain appear? 3; Do you have a cough that produces colored or more abundant sputum than usual? Yes; Do you smoke cigarettes? Yes; Do you have a chronic obstructive pulmonary disease (COPD)? Yes; Do you have a cough? Yes; Have you traveled out of the country in the last 4 weeks? N; Have you noticed a wheezing sound when you exhale? Yes.

Llama3.1-8B:

acute copd exacerbation / infection. \n\nThe answer is based on the patient's history of COPD, smoking, and symptoms of a productive cough, colored or more abundant sputum, and wheezing sound when exhaling.

MedS³

Step 0: Let's break down this problem step by step. \n\nStep 1: Identify key symptoms and risk factors. \n- The patient has a cough producing colored or more abundant sputum than usual. \n- He experiences pain in the lower chest, right side of the chest, right breast, and posterior chest wall (both sides). \n- The pain is sensitive, with an intensity of 2/10 and appeared within 3 days. \n- He has a history of smoking and COPD. \n- He also reports a wheezing sound when exhaling. \n\nStep 2: Consider possible diagnoses based on symptoms and risk factors. \n- Given the patient's history of smoking and COPD, chronic bronchitis or acute exacerbation of COPD could be a possibility. \n- However, the presence of colored sputum and pain in the chest and breast areas could suggest a more complex condition. \n- The patient's symptoms, such as cough, sputum production, and wheezing, are consistent with a respiratory condition. \n\nStep 3: Evaluate the likelihood of each possible diagnosis. \n- Considering the patient's history of smoking and COPD, bronchitis (chronic or acute) is a strong possibility. \n- However, the presence of colored sputum and pain in the chest and breast areas could also suggest bronchiectasis, a condition characterized by damaged airways that can produce excess mucus. \n- Other conditions, such as pneumonia or pulmonary embolism, could also present with similar symptoms, but they are less likely given the patient's history and the absence of other risk factors. \n\nStep 4: Based on the patient's symptoms and risk factors, the most likely diagnosis is bronchitis, likely an acute exacerbation of his chronic COPD. However, considering the presence of colored sputum and pain, bronchiectasis is also a possibility, but given the options provided, **the best answer is bronchitis.**

MedS³-PRM:

Step 0: Let's break down this problem step by step. \n\nStep 1: Identify the key symptoms and risk factors. The patient presents with a cough producing colored or more abundant sputum than usual, pain in the lower chest, right side of the chest, right breast, and posterior chest wall (both sides), which is sensitive and appeared within 3 days. He also reports a history of smoking and COPD. \n\nStep 2: Consider the potential causes of these symptoms. Given the history of smoking and COPD, chronic bronchitis or acute exacerbation of COPD could be a possibility. However, the presence of colored sputum and localized chest pain may suggest a more specific condition. The patient's symptoms could also be consistent with pneumonia or bronchiectasis, but these conditions typically present with more systemic symptoms and findings on physical examination or imaging. \n\nStep 3: Assess the patient's risk factors for specific conditions. The patient's smoking history and COPD increase his risk for chronic obstructive pulmonary diseases (COPDs) and pulmonary infections. However, the patient's age and sex also increase his risk for other conditions such as lung cancer. \n\nStep 4: Based on the patient's symptoms and risk factors, the most likely diagnosis is an acute exacerbation of COPD or chronic bronchitis. However, the presence of colored sputum and localized chest pain also raises the possibility of bronchiectasis. Considering the patient's history of smoking and COPD, the most likely diagnosis among the options provided is bronchiectasis or acute bronchitis/bronchiolitis but I see that the best answer here is an exacerbation of the COPD which is not listed but we do have 'bronchitis' which could be correct in this context. \n\n**The answer is bronchitis.**

Figure 13: Cases of DDXPlus

Question:

Imagine you are a doctor, next I will give you summarization of a patient and please help me to give a suggestion for next treatment. You just need to pick out an answer from ['Intraoperative rad with other rad before/after surgery', 'Intraoperative radiation', 'No radiation and/or cancer-directed surgery', 'Radiation after surgery', 'Radiation before and after surgery', 'Radiation prior to surgery', 'Surgery both before and after radiation']\nAge at diagnosis: 73\n\nRegional nodes positive (1988+): 0\n\nTotal number of in situ/malignant tumors for patient: 1\n\nRadiation recode: None/Unknown\n\nChemotherapy recode: No/Unknown\n\nER Status Recode Breast Cancer (1990+): Positive\n\nPR Status Recode Breast Cancer (1990+): Negative\n\nCS tumor size (2004-2015): 16\n\nDerived HER2 Recode (2010+): Negative\n\nRegional nodes examined (1988+): 8\n\nCOD to site recode: Diseases of Heart\n\nRace recode: White\n\nSex: Female\n\nVital status recode (study cutoff used): Dead\n\nDiagnosis_year: 2010\n\nLast_fu_year: 2015\n\ninterva_years: 5

Llama3.1-8B:

Based on the provided information, I would suggest the following treatment plan:\n\nGiven the patient's age (73) at diagnosis, the presence of a single tumor (1), and the fact that the patient has died from a cause other than breast cancer (Diseases of Heart), it's likely that the cancer was not the primary cause of death.\n\nHowever, considering the patient's ER-positive status and the relatively large tumor size (16), a treatment plan that includes radiation therapy may be beneficial.\n\nGiven the patient's age and the fact that the cancer was not the primary cause of death, I would suggest 'Radiation after surgery' as the next treatment plan. This approach would allow for the removal of the tumor through surgery, followed by radiation therapy to target any remaining cancer cells and reduce the risk of recurrence. \n\nThe answer is 'Radiation after surgery'..

MedS³

Step 0: Let's break down this problem step by step.\n\nStep 1: Identify the key information from the patient's summary. The patient is a 73-year-old female with breast cancer, diagnosed in 2010. The cancer is ER-positive, PR-negative, and HER2-negative. The patient had 8 regional nodes examined, but none were positive. The cancer was 16mm in size.\n\nStep 2: Assess the patient's prognosis and treatment options. Given the patient's age, cancer size, and ER-positive status, the patient is likely to benefit from hormone therapy. However, the patient's PR-negative status suggests that PR-targeted therapy may not be effective.\n\nStep 3: Consider the patient's vital status and last follow-up year. The patient passed away in 2015, which suggests that the cancer may have progressed despite treatment.\n\nStep 4: Evaluate the treatment options for ER-positive breast cancer. The standard treatment for ER-positive breast cancer includes hormone therapy, such as tamoxifen or aromatase inhibitors. However, the patient's PR-negative status and cancer progression suggest that hormone therapy may not be sufficient.\n\nStep 5: Consider the patient's cancer-directed surgery and radiation therapy. The patient's cancer size and regional nodes examined suggest that surgery and radiation therapy may have been used to treat the cancer.\n\nStep 6: Evaluate the radiation therapy options. Given the patient's cancer progression, radiation therapy may have been used to control symptoms or slow disease progression.\n\nStep 7: Consider the patient's overall health and treatment goals. The patient's age and cancer progression suggest that aggressive treatment may not be feasible.\n\nStep 8: Based on the patient's cancer characteristics and treatment history, the most likely treatment approach is radiation after surgery. \n\nThe answer is 'Radiation after surgery'.

MedS³-PRM:

Step 0: Let's break down this problem step by step.\n\nStep 1: Analyze the patient's information to determine the stage and characteristics of the breast cancer. The patient is a 73-year-old female with a tumor size of 16mm, which is likely to be stage I or stage II breast cancer. The patient has estrogen receptor-positive (ER+) breast cancer, but HER2-negative and progesterone receptor-negative (PR-) status. There is no information on lymph node involvement, but 8 nodes were examined.\n\nStep 2: Assess the patient's overall health and treatment goals. The patient is 73 years old, and the cause of death is listed as diseases of the heart. This suggests that the patient may have had underlying cardiovascular conditions that could impact her ability to tolerate certain treatments.\n\nStep 3: Consider the standard treatment options for early-stage breast cancer. For a patient with ER+ breast cancer, hormone therapy is often a recommended treatment option. However, the patient's age and cardiovascular conditions may impact the choice of treatment.\n\nStep 4: Evaluate the radiation therapy options. Given the patient's age and cardiovascular conditions, it may be more appropriate to avoid invasive surgery and radiation therapy. However, the patient's tumor size and lymph node status are not sufficient to rule out the need for radiation therapy entirely.\n\nStep 5: Determine the most appropriate next step in treatment based on the patient's characteristics and overall health. Considering the patient's age, cardiovascular conditions, and ER+ status, a more conservative approach may be preferred. However, there is no clear indication that the patient received adequate radiation therapy. A more conservative option would be to provide hormone therapy and monitor the patient's condition. But among the provided options, \"No radiation and/or cancer-directed surgery\" seems like the best answer given the lack of information on whether surgery was performed and the patient's overall health. \n\nThe answer is 'No radiation and/or cancer-directed surgery'.

Figure 14: Cases of SEER